

TIME: **25** Minutes Total Marks: **20**

Name:

Student_ID:

1. Suppose there is a circle of unknown radius centered at the point $(5, 3)$ in 2D space. Identify all the transformations that would leave both the shape of the circle and the position of the center unchanged. For each of them, provide the corresponding transformation matrix. If any transformation involves multiple stages, mention that too. 7
2. An object undergoes a 3D rotation. The rotation is such that it leaves the X-axis unchanged, but maps the positive Y-axis to the negative Z-axis. Determine the rotation matrix that would reverse this transformation and restore the object to its original orientation. 3

3. You are given the following 4×4 transformation matrix in a 3D space:

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$$M = \begin{bmatrix} 0.866 & 0 & 0.5 & 6.928 \\ 0 & 1 & 0 & 4 \\ -0.5 & 0 & 0.866 & 0.464 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

This matrix results from a single rotation about one of the principal axes combined with a translation. However, the order of these transformations is unknown. All we know is that the translation amount along the X-axis is less than 6.

- a. Determine the rotation angle (in degrees) and the rotation axis.
- b. Identify the order of operations and find the translation amounts in the X, Y, and Z directions.